

LO2: Reconfiguring Digital Value Chains & Business Model Canvas

The four key shifts shown in the slide are:

1. **Linear → Ecosystem Value Chains**
2. **Physical Resources → Digital Key Resources**
3. **High CapEx → Cloud OpEx**
4. **Traditional Revenue → Digital Revenue Streams**

1. Linear → Ecosystem Value Chains

What does it mean?

A **traditional linear value chain** usually looks like:

Supplier → Business → Distributor → Customer

The business controls most of the process.

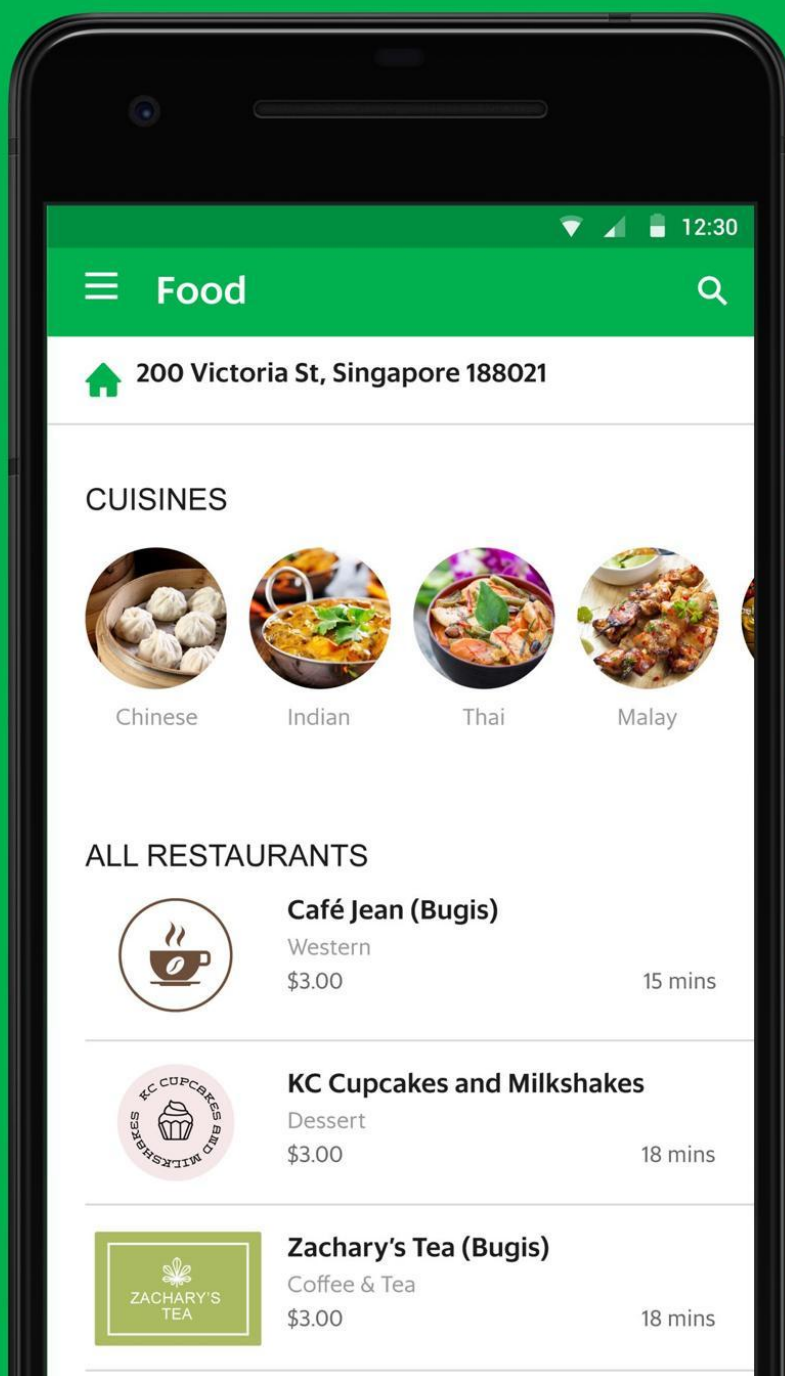
A **digital ecosystem** connects multiple businesses, customers, platforms and technology providers.

Business ↔ Customers ↔ Partners ↔ Platforms ↔ Suppliers ↔ Technology Providers

Instead of doing everything itself, the company uses a digital platform to coordinate different participants.

Example: Grab

**Browse all kinds of
food near you!**







Traditional taxi business:

Taxi company → Taxi → Passenger

A digital ecosystem such as Grab is much more complex:

Customer



Grab platform



Driver | Restaurant | Delivery Partner



Payment provider | Map provider | Technology providers

What is happening?

A customer uses the Grab app to:

- Book a ride
- Order food
- Make payment

- Track delivery
- Rate the service

Grab does not need to own every restaurant, car or delivery vehicle.

Instead, the **digital platform connects different participants**.

Business Model Canvas connection

BMC Component	Traditional Taxi	Digital Ecosystem
Key Partners	Taxi operators	Drivers, restaurants, payment providers, technology partners
Key Activities	Provide taxi service	Operate platform, matching, payments, data management
Key Resources	Cars, drivers	App, platform, data, algorithms
Channels	Taxi stand/phone	Mobile app
Customer Relationships	Human interaction	App + automated recommendations/support

Student takeaway

Digital ecosystems allow a business to create value by connecting multiple participants rather than owning every resource involved in delivering the product or service.

2. Key Resources: Physical Assets → Data, AI, Cloud & Algorithms

What does it mean?

Traditional businesses often depend heavily on **physical assets**.

For example:

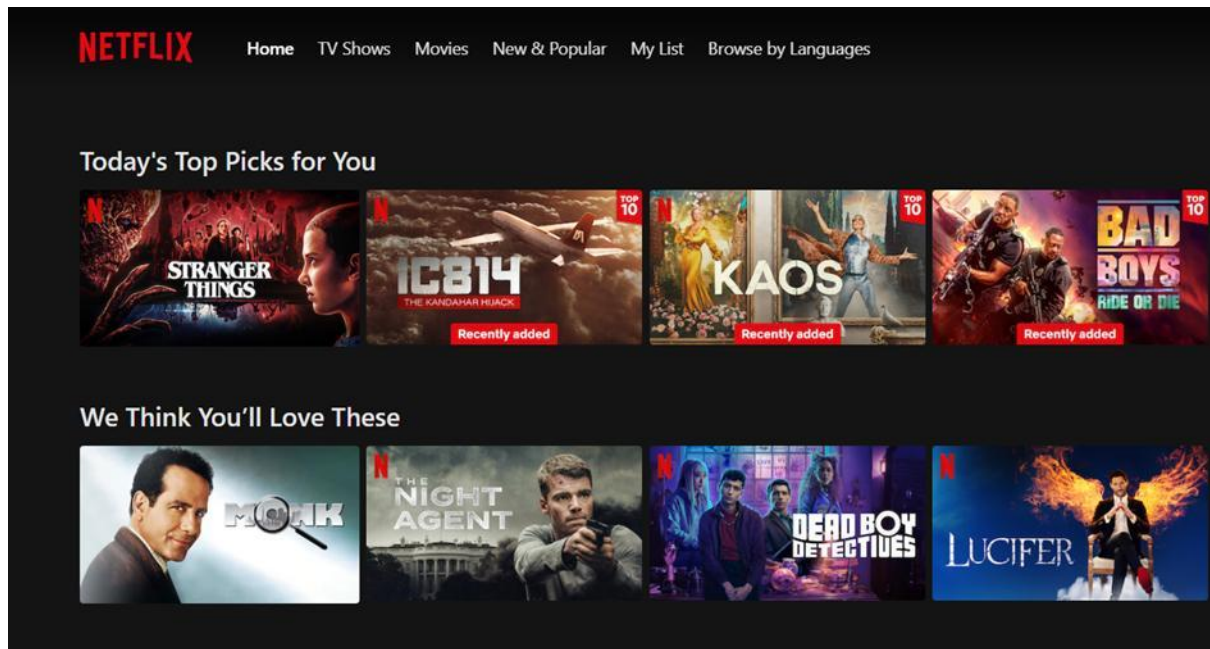
Hotel → Buildings + Rooms + Furniture + Reception

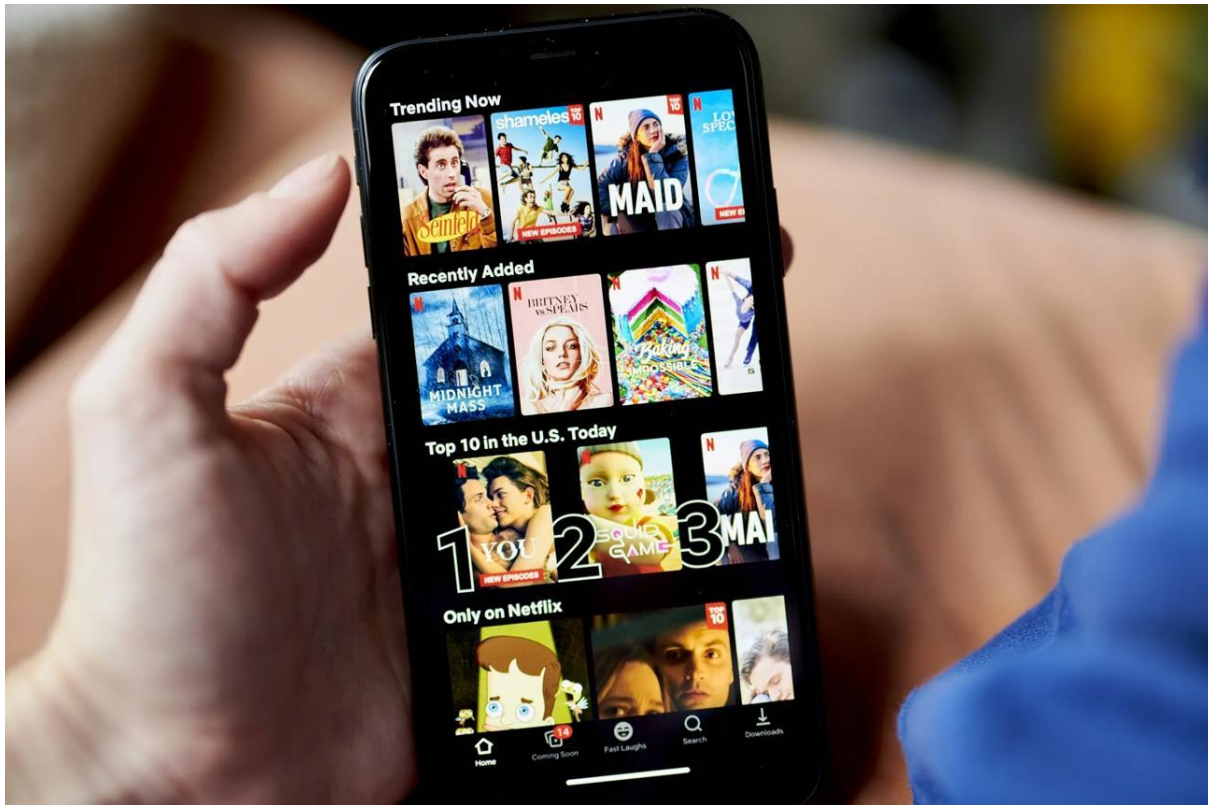
Digital businesses can create significant value from **intangible digital resources**:

- Customer data
- Software
- Algorithms
- AI models

- Cloud infrastructure
- Digital platforms
- Intellectual property

Example: Netflix





A traditional video rental business needs:

- Physical stores
- DVDs
- Shelves
- Retail employees
- Storage space

Netflix's key resources are much more digital:

Netflix's digital resources

1. Customer data

Netflix can analyse:

- What users watch
- How long they watch
- What they search for
- When they stop watching

- Which genres they prefer

2. Recommendation algorithms

The system uses viewing behaviour to recommend content.

For example:

"Because you watched X, you might also like Y."

3. Cloud and streaming infrastructure

Customers don't need to visit a physical store.

4. Digital platform

The Netflix application becomes the main channel through which customers access the service.

The important shift

Traditional video rental:

Physical stores + DVDs = major resources

Digital streaming:

Data + software + algorithms + content + cloud infrastructure = major resources

Business Model Canvas

The **Key Resources** box changes dramatically.

Instead of:

Stores + DVD inventory + physical equipment

It becomes:

Platform + customer data + recommendation algorithms + content library + cloud infrastructure

3. Cost Structure: High CapEx → Cloud OpEx

This is one of the more technical concepts on the slide.

What is CapEx?

CapEx = Capital Expenditure

Money spent on major physical assets that are expected to be used over a long period.

Examples:

- Servers

- Buildings
- Data centres
- Machinery
- Vehicles

Traditional IT example

Suppose a company wants to build its own data centre.

It may need to spend:

\$2 million on servers + networking equipment + cooling + physical infrastructure.

This is a large **upfront capital investment**.

What is OpEx?

OpEx = Operating Expenditure

Ongoing expenses required to operate the business.

Examples:

- Cloud computing fees
- Software subscriptions
- Internet services
- Employee salaries
- Monthly platform fees

Instead of purchasing its own servers, a company can use cloud providers and pay according to usage.

Example: A Small Online Retailer

Imagine **ABC Fashion** wants to launch an online store.

Traditional approach

ABC Fashion purchases:

- Physical servers
- Networking equipment
- Storage
- Backup systems

It spends a large amount before getting many customers.

Cloud approach

ABC Fashion uses cloud computing.

It pays based on its usage.

For example:

Low customer traffic → lower computing requirements

High customer traffic → scale computing resources upward

So the cost structure changes from:

Large upfront investment



More flexible ongoing operating costs

Real-world example: Shopify merchants

A small business doesn't need to build its own e-commerce infrastructure.

It can use an existing digital commerce platform and pay subscription/transaction-related fees.

This dramatically reduces the amount of infrastructure the business needs to purchase itself.

Business Model Canvas connection

The **Cost Structure** changes from:

Buildings + servers + IT infrastructure + maintenance

to:

Cloud services + software subscriptions + platform fees + digital marketing + transaction fees

Important clarification

The phrase "**near-zero marginal cost**" on the slide does **not** mean digital businesses have zero costs.

For example, streaming one additional video to a customer still consumes computing, bandwidth and other resources.

It means that **the cost of serving one additional customer can be relatively small compared with traditional physical expansion**, particularly after the digital infrastructure has been established.

4. Revenue Streams: Traditional → Subscription, Freemium, Marketplace & Data

Digital businesses have created many new ways to generate revenue.

The slide highlights four important examples.

A. Subscription — SaaS

SaaS = Software as a Service

Instead of selling software once, the company charges customers regularly.

Example: Microsoft 365

Traditional software model:

Buy software → Install → Use

Digital SaaS model:

Subscribe → Access online → Pay monthly/yearly

For example, a business may pay for Microsoft 365 based on the number of users and the selected plan.

Business Model Canvas

Revenue Stream:

Recurring subscription revenue

Why is this attractive?

The business can potentially create:

- Predictable recurring revenue
- Long-term customer relationships
- Continuous opportunities for upgrades

B. Freemium

Freemium = Free + Premium

The basic service is free, but customers pay for additional features.

Example: Spotify

9:41



My playlist



Mattias Johansson

25 likes • 2h 26m



Add song



Shadows

Aur Revoir Simone



4ever

Clairo



Freak Scene

Dinosaur Jr.



Velvet Ring

Big Thief



Home



Search



Your Library

Pick your Premium

We've got a plan that's right for you.

Subscribe

30 days free

Spotify Premium

Subscribe for ₹119.00/month after trial.

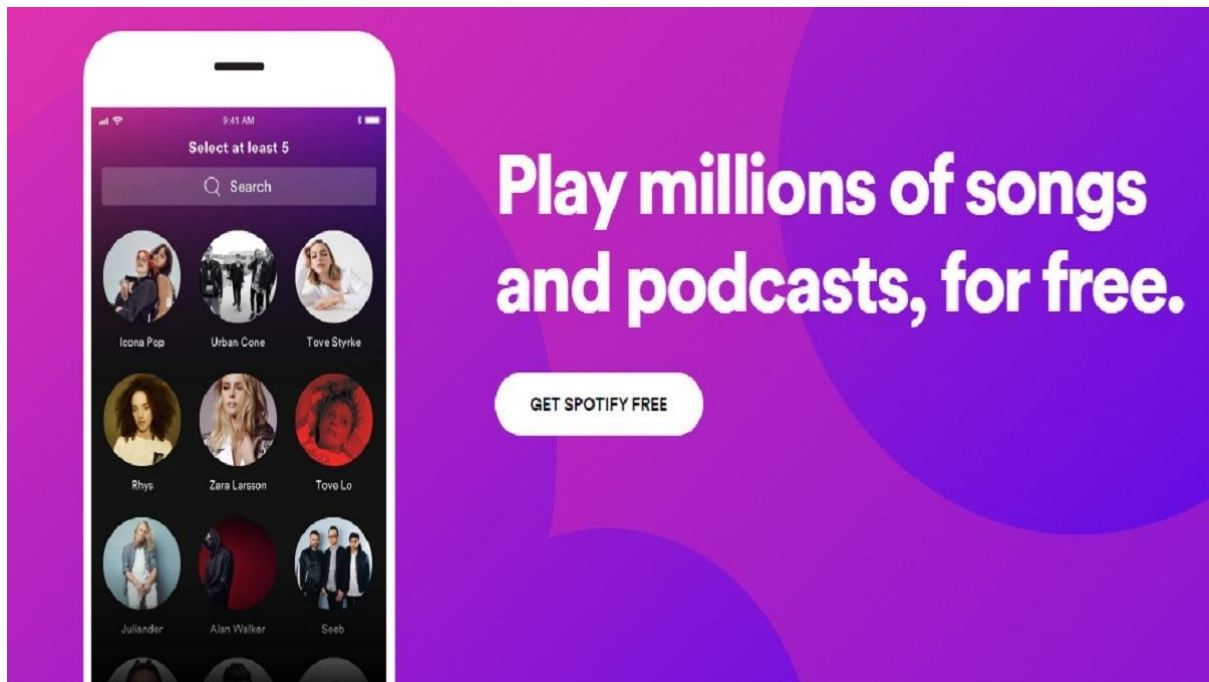
Cancel anytime.

VISA



GET 30 DAYS FREE

* Terms and conditions apply.



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Spotify can attract users with a free service.

Free users

May receive:

- Advertising
- Restrictions on certain features
- Limited control over listening

Premium users

Pay for additional benefits such as:

- Ad-free listening
- Additional control
- Offline listening

So the business model is:

Free users → attract large audience → some users convert to Premium → subscription revenue

C. Marketplace Transaction Take-Rate

This is particularly important for businesses such as **Grab, Airbnb and online marketplaces**.

The platform doesn't necessarily own the products being sold.

Instead:

Customer → Platform → Seller/service provider

The platform takes a percentage or fee from the transaction.

Example: GrabFood

Imagine a customer orders:

\$30 meal

The restaurant prepares the food.

The platform facilitates:

- Customer discovery
- Ordering
- Payment
- Delivery coordination
- Tracking

The platform may earn revenue through commissions and/or fees associated with the transaction.

Key idea

The platform can generate revenue **without producing the food itself**.

This is an excellent example of a digital ecosystem.

D. Data Monetisation

Data can become an important **digital asset**, although businesses must comply with applicable privacy and data-protection laws.

A business can create economic value from data through activities such as:

- Analytics
- Personalisation
- Advertising
- Business intelligence
- Insights for partners

Example: E-commerce platform

Suppose an online retailer knows:

- Which products customers search for
- Which products are frequently purchased
- Which products are abandoned in carts
- Which promotions work
- Which customer segments buy particular products

The business can use these insights to:

Improve recommendations → Increase conversion → Increase revenue

A platform may also provide **aggregated or appropriately governed insights** to business partners.

Putting All Four Shifts Together

Let's use **Grab** to connect the concepts.







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Traditional Business Model Digital Business Model

Taxi company owns/controls vehicles	Platform connects drivers and customers
Linear value chain	Ecosystem
Physical assets are critical	App, data and algorithms are critical
Large physical infrastructure	Cloud-based infrastructure
Customer calls/bookings	Mobile app
Fixed service relationship	Data-driven personalised experience
Payment for individual service	Transaction fees, commissions, subscriptions and other digital revenue mechanisms
Limited customer information	Rich customer and transaction data

This demonstrates how **one digital transformation can affect almost every part of the Business Model Canvas.**

Another Excellent Example: Airbnb

Airbnb is particularly useful for teaching this topic because students can immediately see the difference between a traditional hotel and a digital platform.

Traditional hotel

Hotel owns building → Provides rooms → Customer pays hotel

Airbnb

Host → Airbnb platform → Guest

Airbnb doesn't need to own most of the properties listed on its platform.

1. Ecosystem Value Chain

Airbnb connects:

Hosts ↔ Airbnb ↔ Guests

and also interacts with:

- Payment providers
- Cleaning/service providers
- Technology providers
- Local businesses

2. Digital Key Resources

Airbnb's key resources include:

- Digital platform
- Customer/host data
- Search and recommendation algorithms
- Reviews and ratings
- Brand
- Payment infrastructure

3. Cloud OpEx

Rather than each property needing its own digital booking infrastructure, Airbnb operates a centralised digital platform supported by scalable technology infrastructure.

4. Revenue

Airbnb generates revenue primarily through **fees associated with bookings** rather than owning the properties being rented.

A Simple Classroom Explanation

You can explain the entire slide to students using this scenario:

Traditional Pizza Business

Imagine a pizza company.

Old model:

Buy shop → Buy ovens → Hire workers → Make pizza → Customer visits → Customer pays

Now digitise it.

Digital Pizza Business

Step 1 — Ecosystem

Restaurant + Delivery platform + Drivers + Payment provider + Customer

Step 2 — Digital resources

Instead of relying only on ovens and restaurants:

Customer data + ordering app + algorithms + digital payment + CRM

Step 3 — Cloud

Instead of building expensive IT infrastructure:

Use cloud-based POS, CRM and ordering systems.

Step 4 — Digital revenue

Instead of only:

Customer buys pizza → \$20

The business can introduce:

Pizza sales + delivery fees + subscriptions + loyalty programme + marketplace commissions

How This Connects to the Business Model Canvas

This is probably the **most important connection for LO2**.

BMC Component	Digital Transformation
Key Partners	Ecosystem partners become more important
Key Activities	Platform management, data analytics, automation
Key Resources	Data, AI, software, algorithms, cloud
Value Proposition	Convenience, personalisation, speed, accessibility
Customer Relationships	Automated and data-driven relationships
Channels	Apps, websites, social media, digital platforms
Customer Segments	More precise data-driven segmentation
Cost Structure	Cloud OpEx, software subscriptions, digital marketing
Revenue Streams	Subscription, freemium, commissions, transaction fees, advertising

In one sentence:

Digital transformation reconfigures the Business Model Canvas by changing who participates in value creation, what resources are important, how the business operates and how it earns revenue.

Easy Student Framework

For an assessment or classroom activity, give students a traditional business and ask them to answer these **four questions**:

① **Ecosystem**

Who can we connect with digitally?

Suppliers + partners + customers + platforms

② **Digital Resources**

What digital assets could create value?

Data + AI + algorithms + software + cloud

③ **Cost Structure**

What physical infrastructure can technology reduce or replace?

Servers → Cloud

Physical paperwork → Digital systems

Physical stores → E-commerce

④ **Revenue**

How can digital technology create new revenue?

Subscription

Freemium

Commission

Transaction fee

Advertising

Data-driven services

That gives students a practical way to move from understanding the theory to actually "reconfiguring" a Business Model Canvas.